# Counterfactual Explanations and Algorithmic Recourse for Trustworthy AI

Research Seminar

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2026-03-11

# The Ground Truth (Reality)

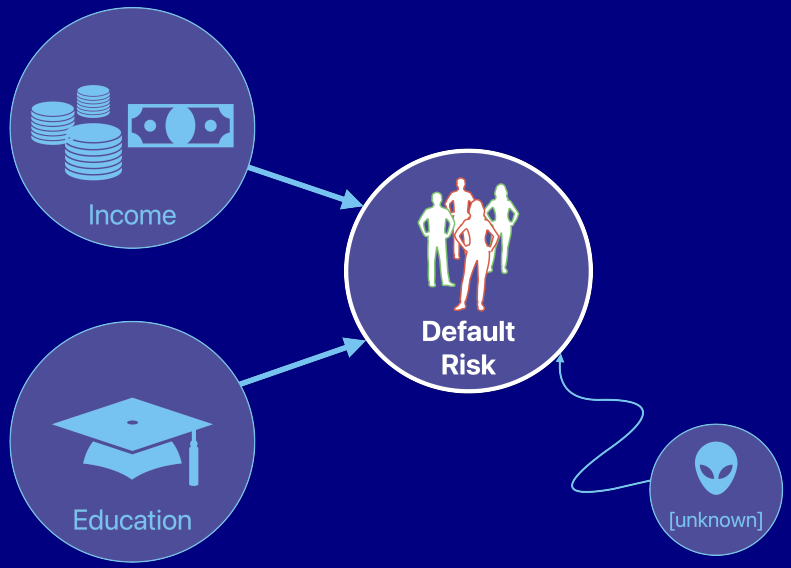



Figure 1: Predictors of default risk.

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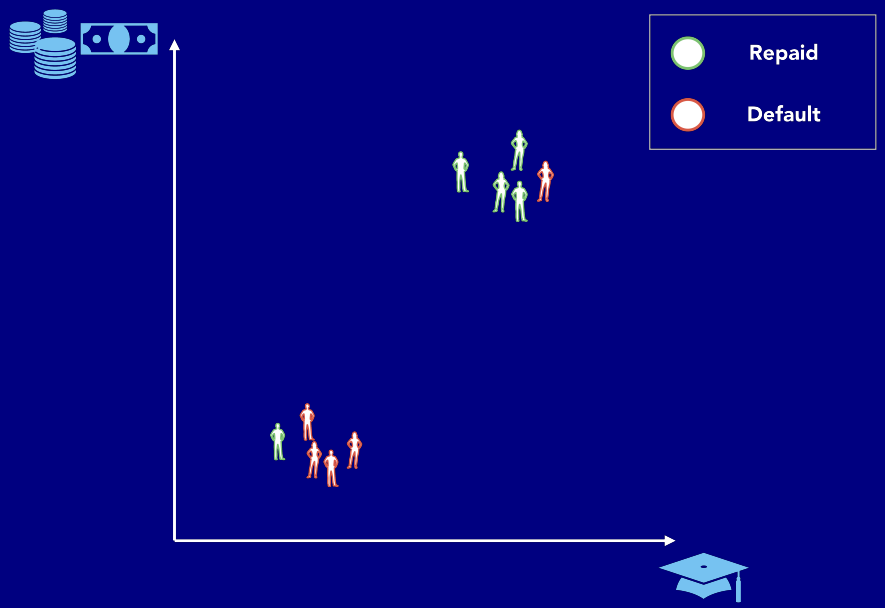



Figure 2: Ground truth outcomes across two predictors.

# Black-Box AI

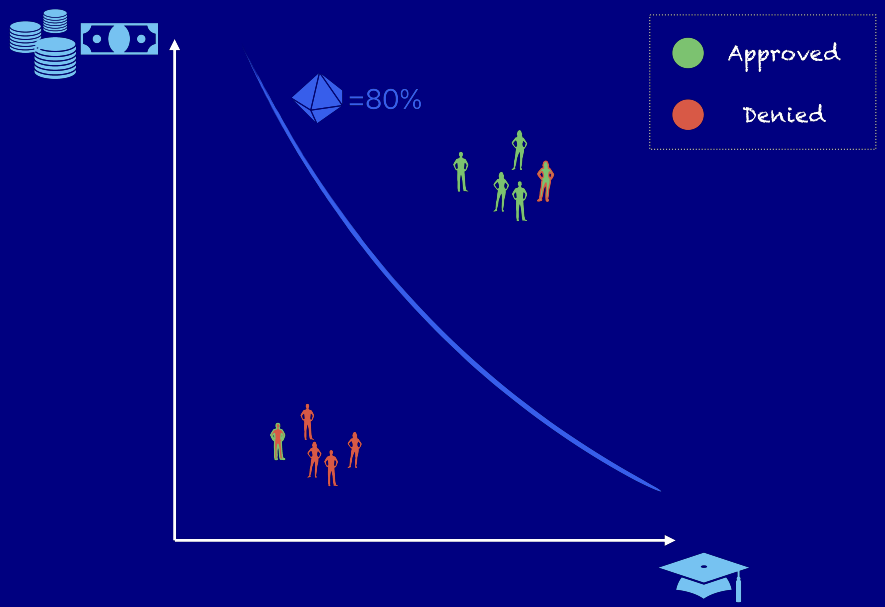



Figure 3: Classifier predicts correctly 8 out of 10 times.

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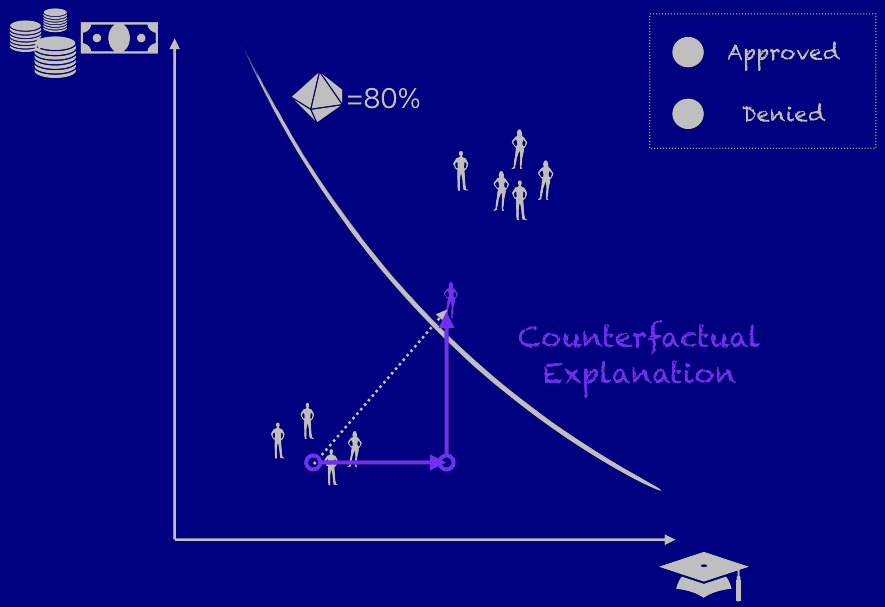



Figure 4: Simple counterfactual explanation for the black-box AI. JuliaCon Proceedings 2023

# Black-Box AI

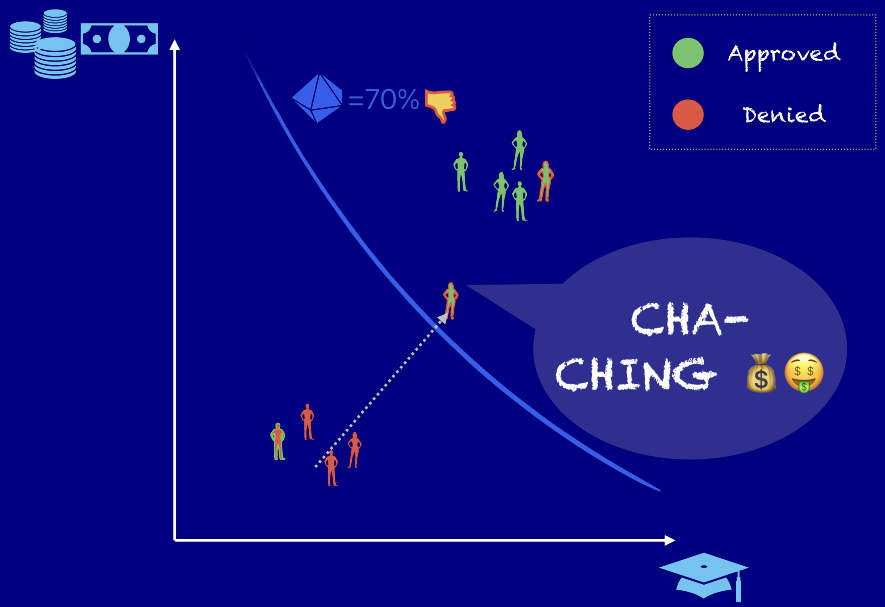



Figure 5: One happy recourse recipient, many losers. IEEE SaTML 2023

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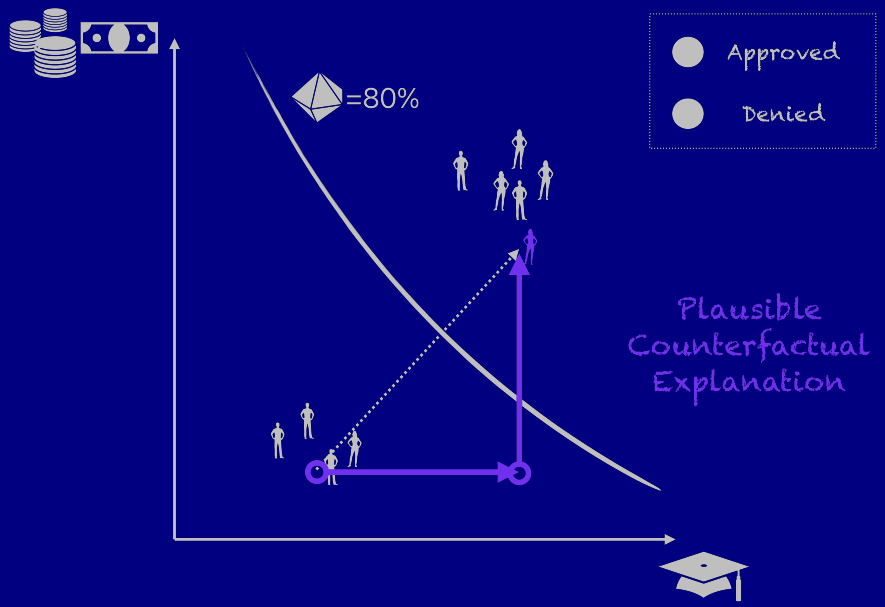



Figure 6: Plausible counterfactual explanations for the black-box AI. IEEE SaTML 2023

# Black-Box AI

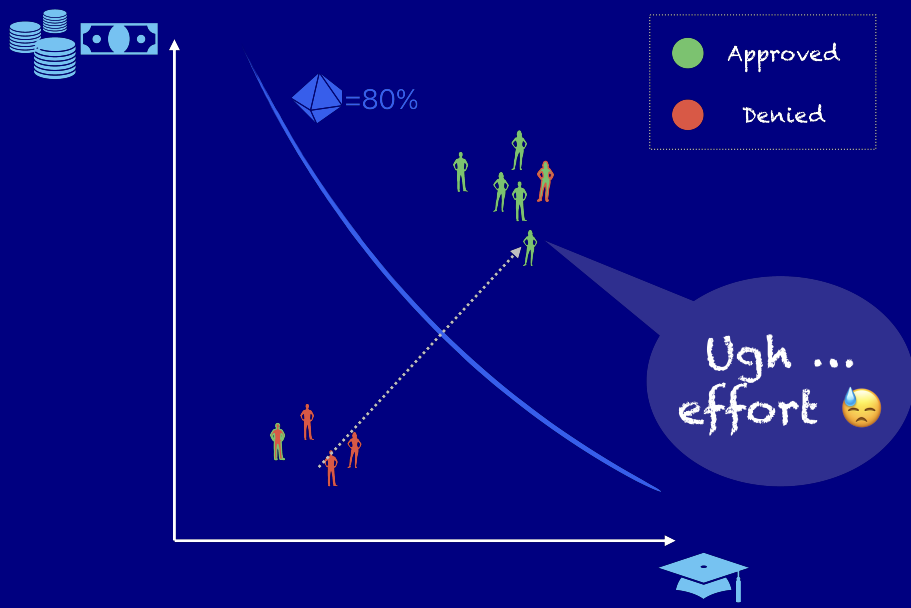



Figure 7: One somewhat happy recourse recipient, no losers. IEEE SaTML 2023

# Big, Beautiful Black-Box AI

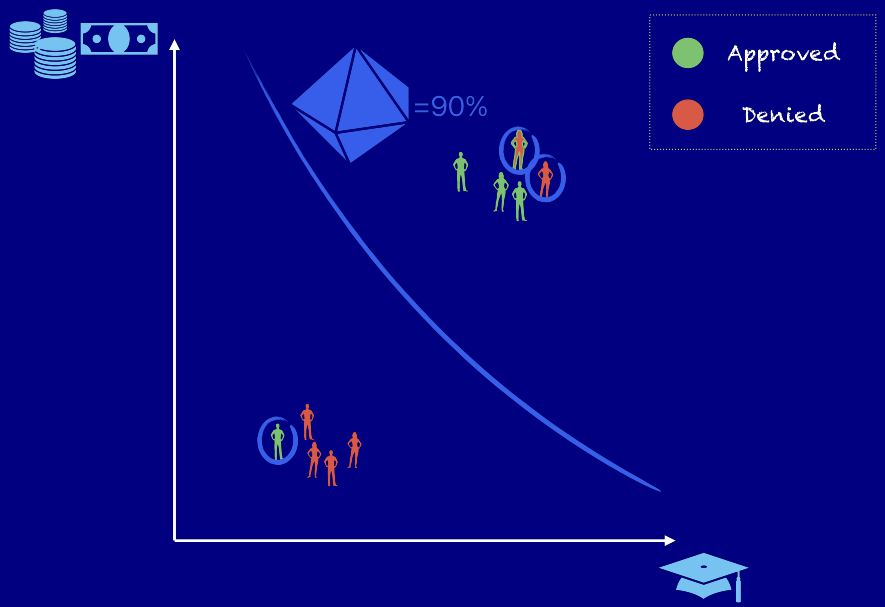



Figure 8: Classifier predicts correctly 9 out of 10 times. But ... AAAI 2024

# Big, Beautiful Black-Box AI

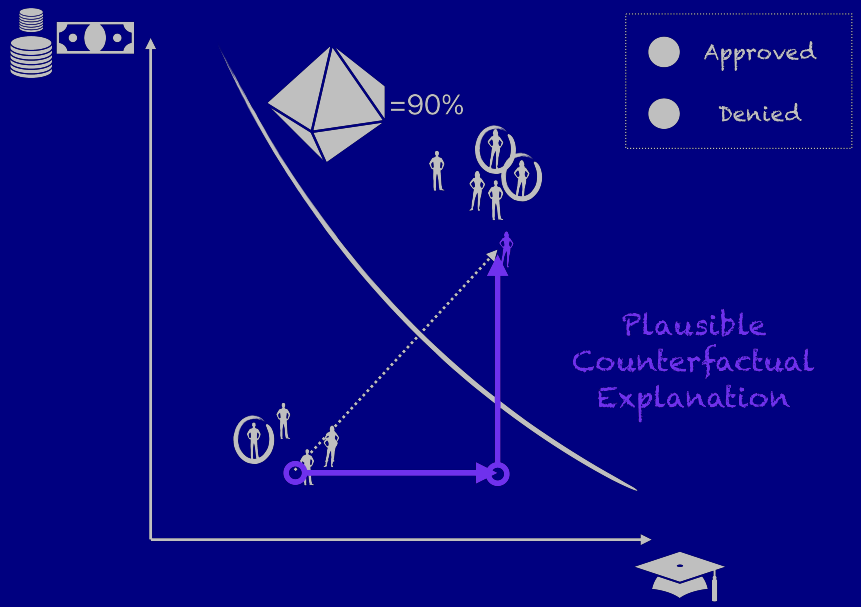



Figure 9: Plausible counterfactual explanations remains valid. Happy days? AAAI 2024

# Big, Beautiful Black-Box AI

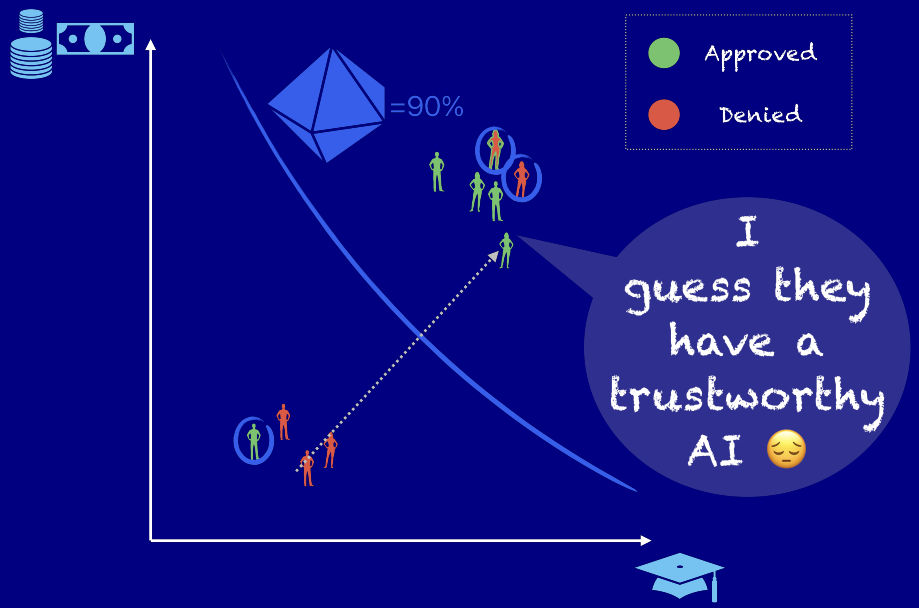



Figure 10: White-washed black-box: plausible CE hides bias. AAAI 2024

# Holding Models Accountable

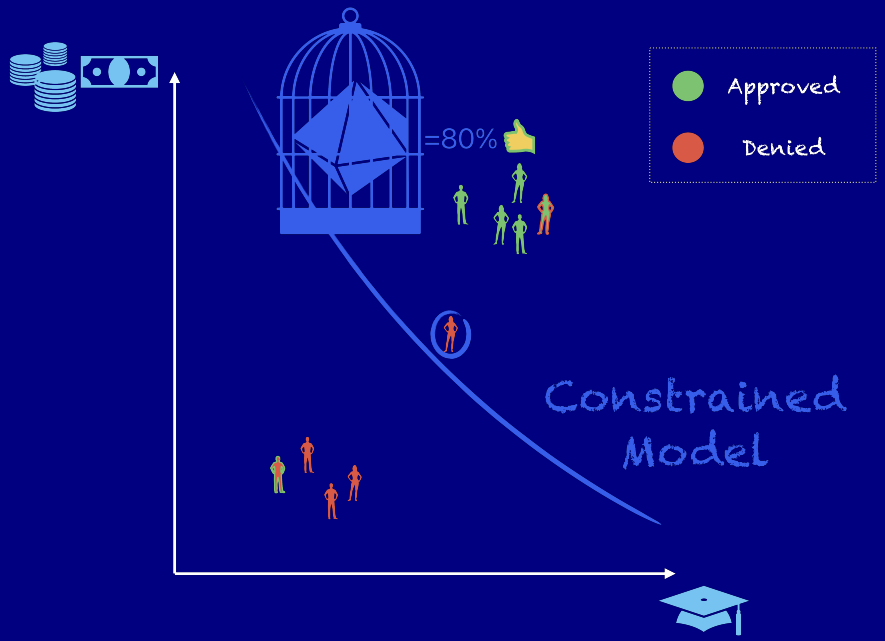



Figure 11: A model trained to use plausible explanations for predictions. IEEE SaTML 2026

# ‘ok but agi bruh’

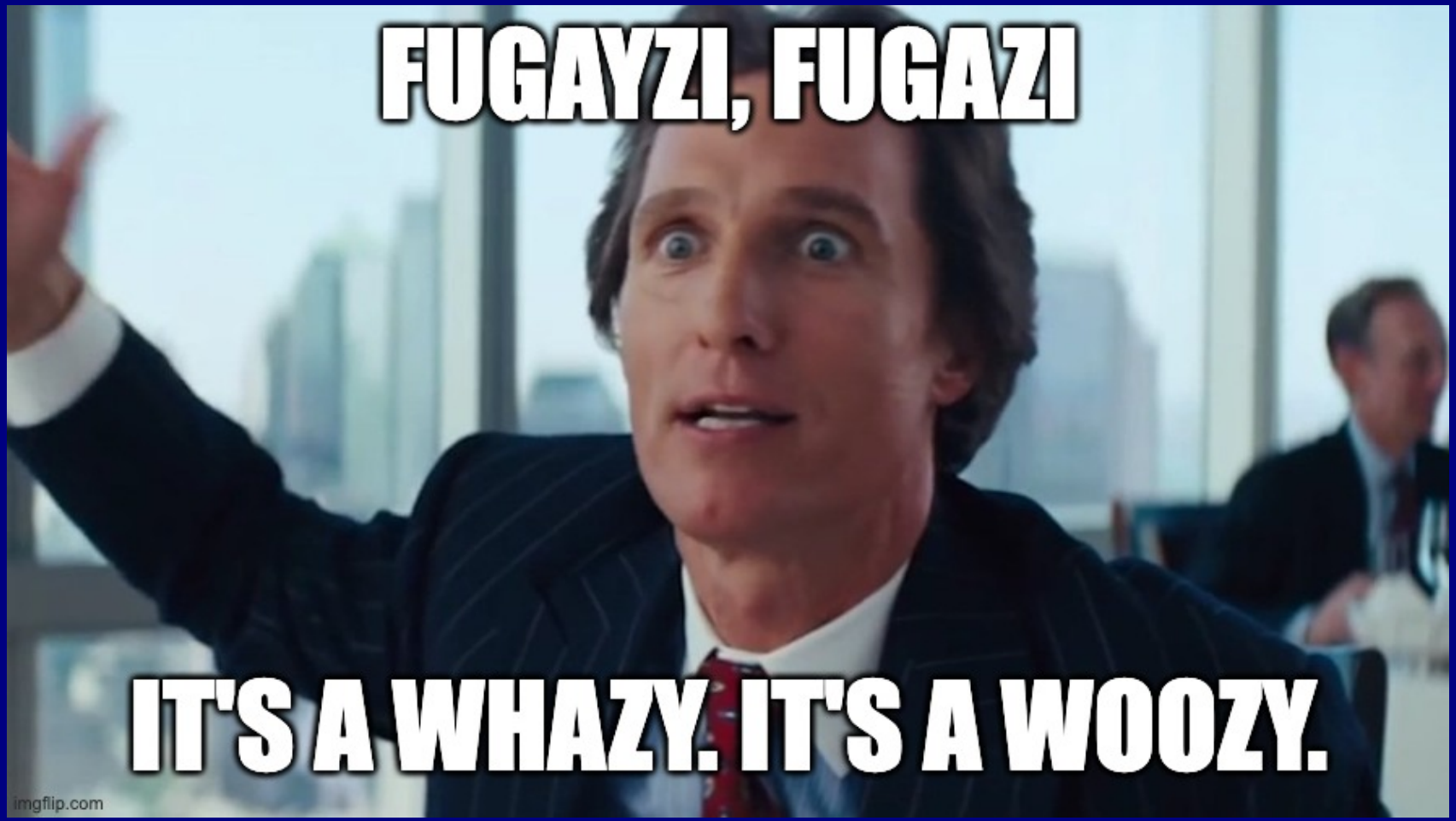



Figure 12: My personal take on “AGI by 2027”. ICML 2024

# In all seriousness ...

- Useful? Absolutely

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- AGI? Sentient? Conscious? No: 'emergence' in complex systems does not hint at any of this

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  - Example 2: tornado -> just dust and debris, but also a possible disaster
- Does it matter?